# Weighted Averages

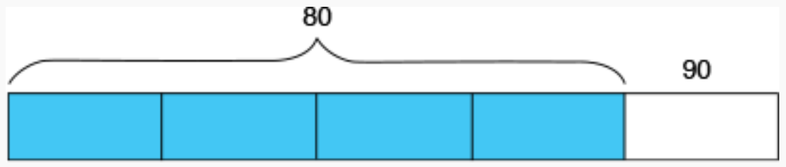
Sometimes, the values in a data set may not be equally important. For example, a final exam might count as two tests. If you were finding your average in the class, your final exam score would be worth more than your other scores.

When some values in a data set are worth more than others, you must find the **weighted average**.

## Example 1

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Aparna's average for the first 4 tests is 80. Her score on the fifth test is 90. What is her average in the class? You might be tempted to say 85 because it is the mean of 80 and 90. Look at the diagram below. Note that 80 represents 4 tests, but 90 represents only 1. You must find the weighted average.




**Step 1**

Multiply each score by the weight. Then add.

$(80 \times 4) + (90 \times 1) = 410$

**Step 2**

Divide by the number of values. Aparna took 5 tests.

$410 \div 5 = 82$

Aparna's average is **82**.

Some averages are calculated by assigning percentages to different categories. These percentages must add up to 100%. Since the percentages represent one whole, you don't need to divide by the number of values.

## Example 2

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In Ms. Padilla's class, tests are worth 30% of the final grade, homework is worth 20%, participation is worth 10%, and the final exam is worth 40%. Jeff earns an 80 on the tests, 95 on homework, 100 on participation, and an 85 on the final exam. What is his average in the class?

**Step 1**

Multiply Jeff's scores by the percent weights.

Tests: $80 \times 0.3 = 24$
Homework: $95 \times 0.2 = 19$
Participation: $100 \times 0.1 = 10$
Final Exam: $85 \times 0.4 = 34$

**Step 2**

Add.

$24 + 19 + 10 + 34 = 87$

Jeff has an average of **87** in the class.

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